

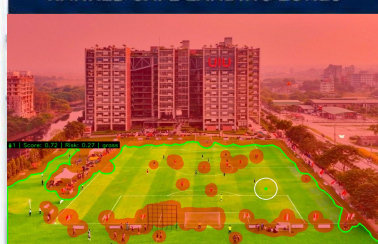
# SAFE LANDING ZONE DETECTION FOR UAVS EMERGENCY LANDING USING SEMANTIC SEGMENTATION

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RANKED SAFE LANDING ZONES



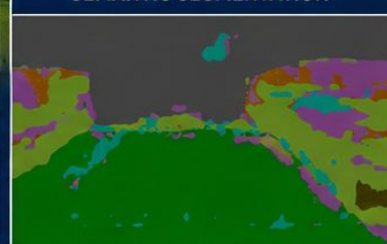
Strongest Rank 2-5 Lower-ranked

LANDING SUITABILITY / RISK MAP



High Risk Medium Risk Low Risk

SEMANTIC SEGMENTATION

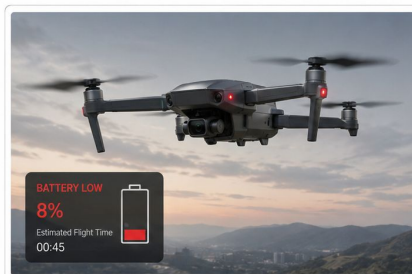


Grass Tree Water Building Road/Other

# Introduction

Autonomous UAVs require safe emergency landing during:

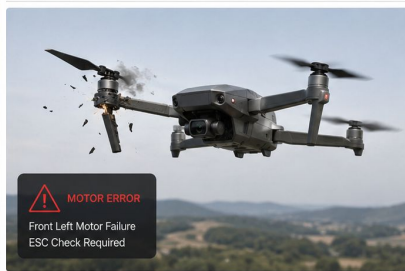
- Battery depletion
- Communication loss
- Hardware failure
- Unexpected environmental risk



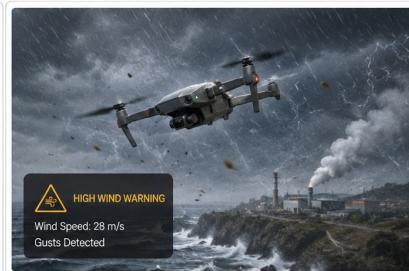
Battery Depletion



Communication Loss



Hardware Failure

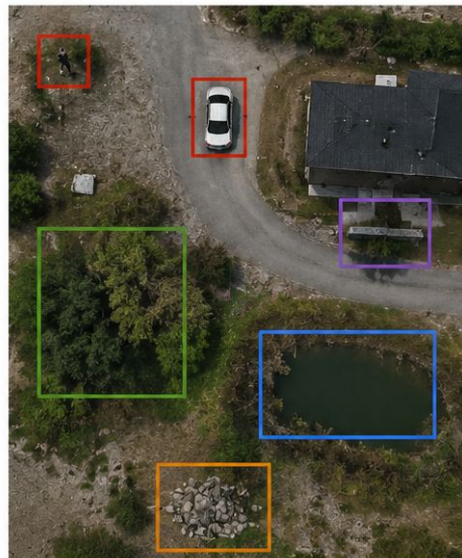
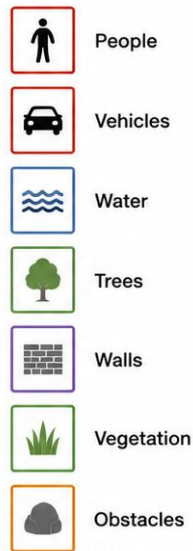


Unexpected Environmental Risk

# Motivation

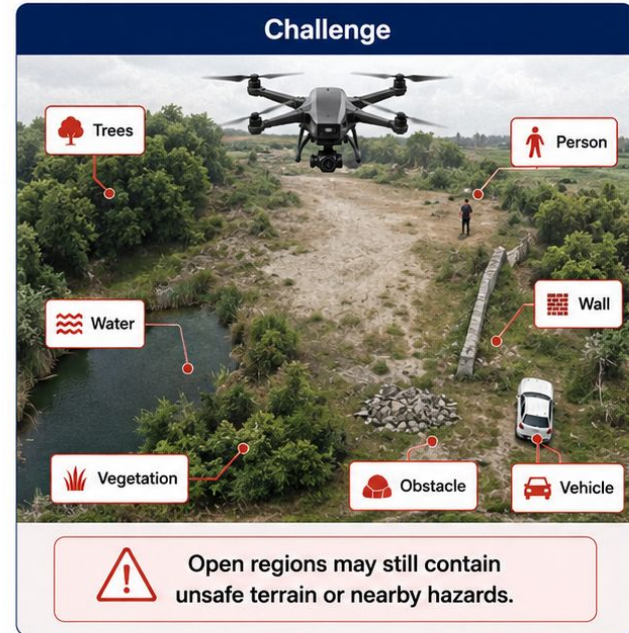
- Manual landing is difficult in emergency situations
- GPS-only navigation cannot detect local hazards such as:

- People
- Vehicles
- Water
- Trees
- Walls
- Vegetation
- Obstacles



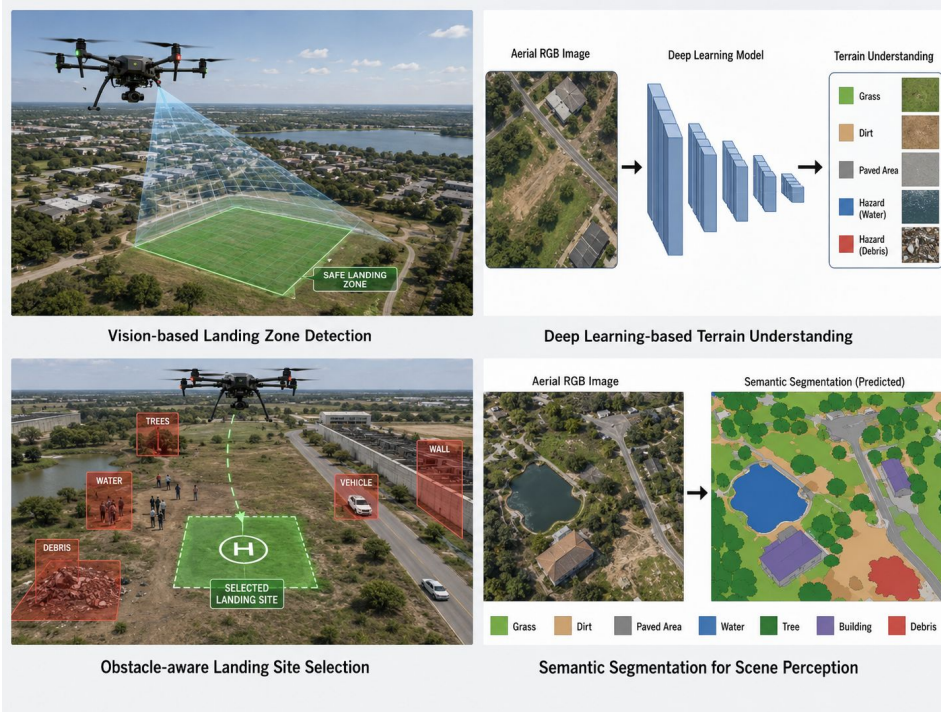
# Problem Statement

- Detecting a safe landing area from aerial images is challenging because open regions may still contain unsafe terrain or nearby hazards.



# Related Works

Recent UAV landing research focuses on:



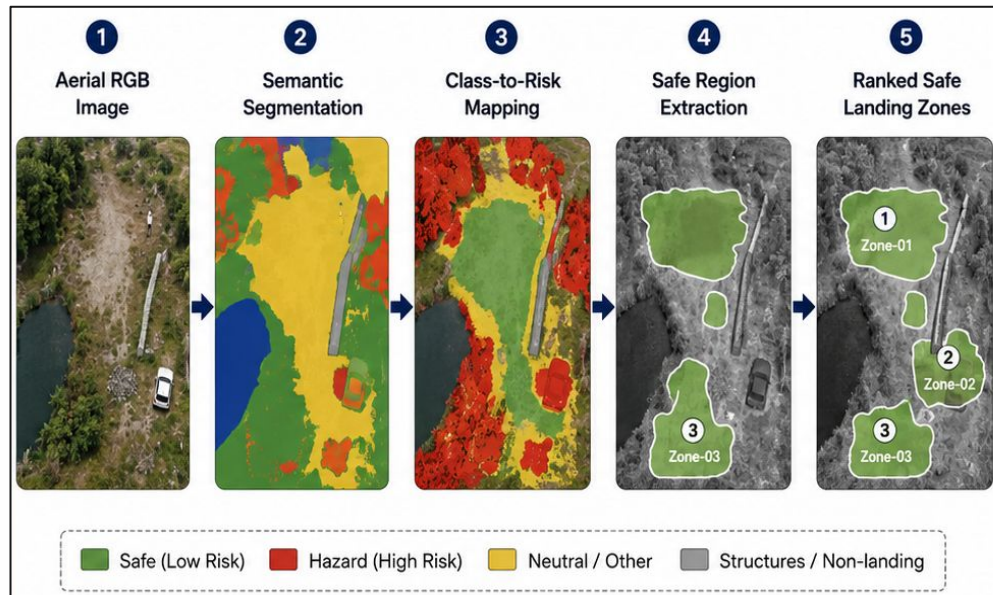
# Key Gaps

- **Limited RGB-based landing safety analysis**  
Many systems depend on LiDAR, RGB-D, or multi-sensor setups, while RGB-only explainable methods are less explored.
- **Weak hazard-aware safe-zone filtering**  
Open regions may still contain nearby hazards such as water, trees, vehicles, people, or obstacles.
- **Lack of clear candidate ranking**  
Existing approaches often detect possible landing areas but do not clearly rank them using area, safety margin, and hazard distance.

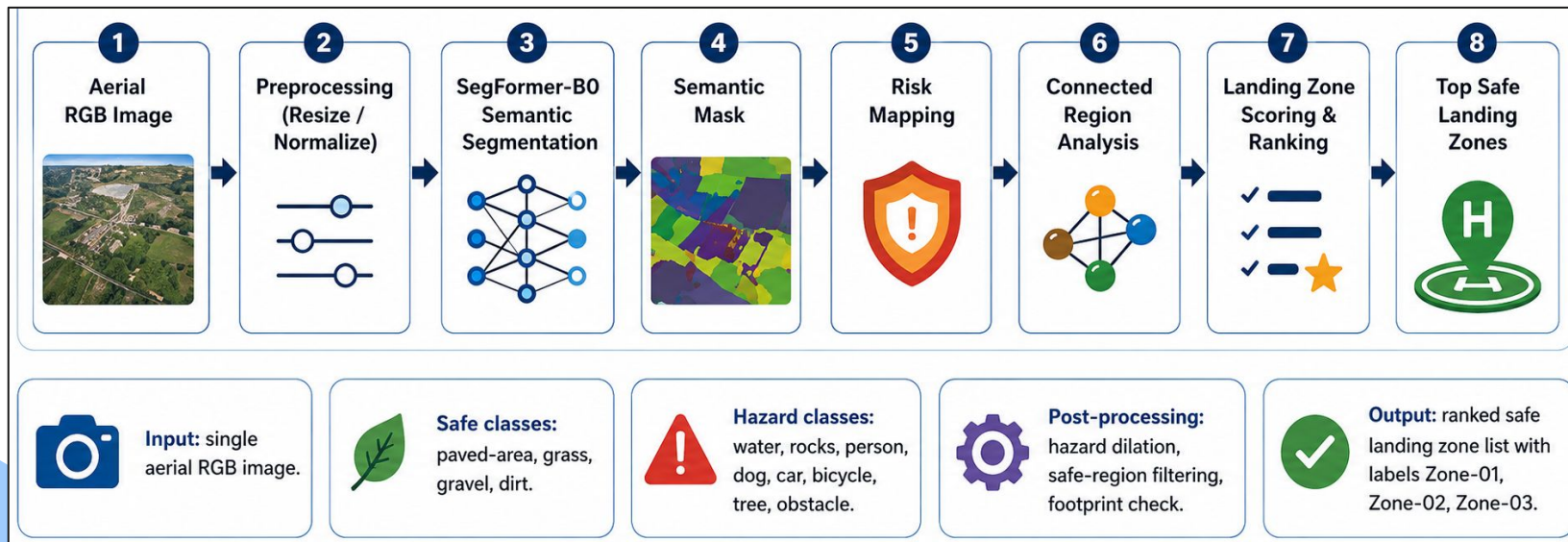
# Objective

To develop an RGB image-based UAV emergency landing framework that:

1. Segments aerial scenes
2. Classifies terrain into safe and unsafe regions
3. Rejects hazard-dominated areas
4. Extracts candidate landing zones
5. Ranks safe zones using measurable criteria



# Proposed Framework



# Dataset and Model

## Dataset

Semantic Drone Dataset (sample images)



Image A

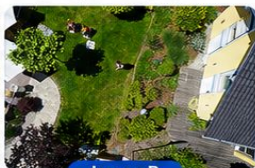


Image B

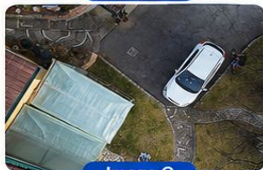
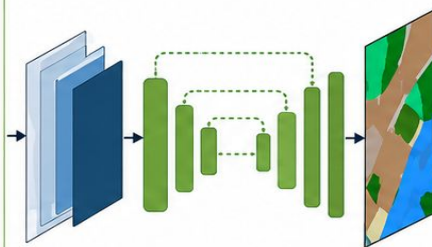


Image C



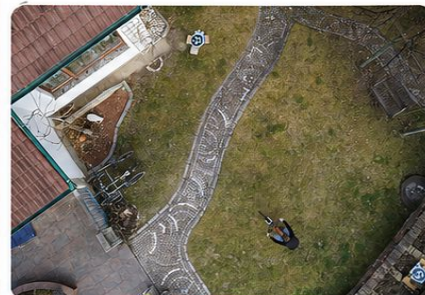
Image D

## Model



SegFormer-B0 semantic segmentation model

## Input



One aerial RGB image

## Main Classes



Paved-area



Dirt



Grass



Gravel



Water



Vegetation



Person



Car



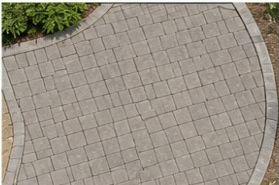
Tree



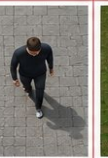
Obstacle

# Risk Mapping Criteria

## SAFE SURFACE CLASSES (Suitable for Landing)

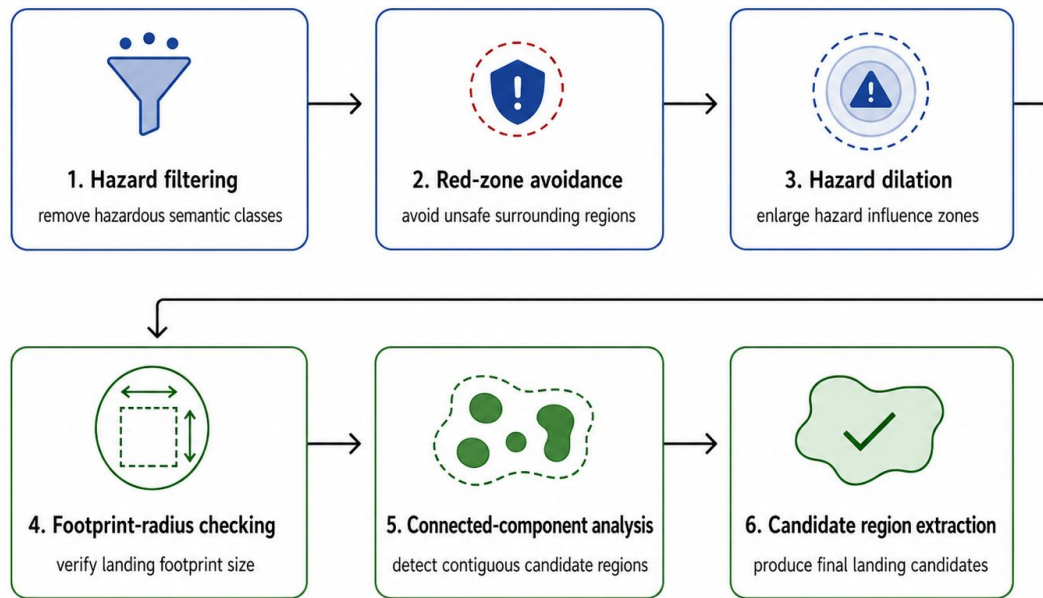
| Class            | Paved-area                                                                        | Dirt                                                                               | Grass                                                                               | Gravel                                                                              |
|------------------|-----------------------------------------------------------------------------------|------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|
| Sample Image     |  |  |  |  |
| Risk Level       | LOW                                                                               | LOW                                                                                | LOW                                                                                 | LOW                                                                                 |
| Risk Score (0–1) | 0.00 – 0.20                                                                       | 0.10 – 0.30                                                                        | 0.00 – 0.25                                                                         | 0.10 – 0.30                                                                         |

## HAZARD CLASSES (Unsafe for Landing)

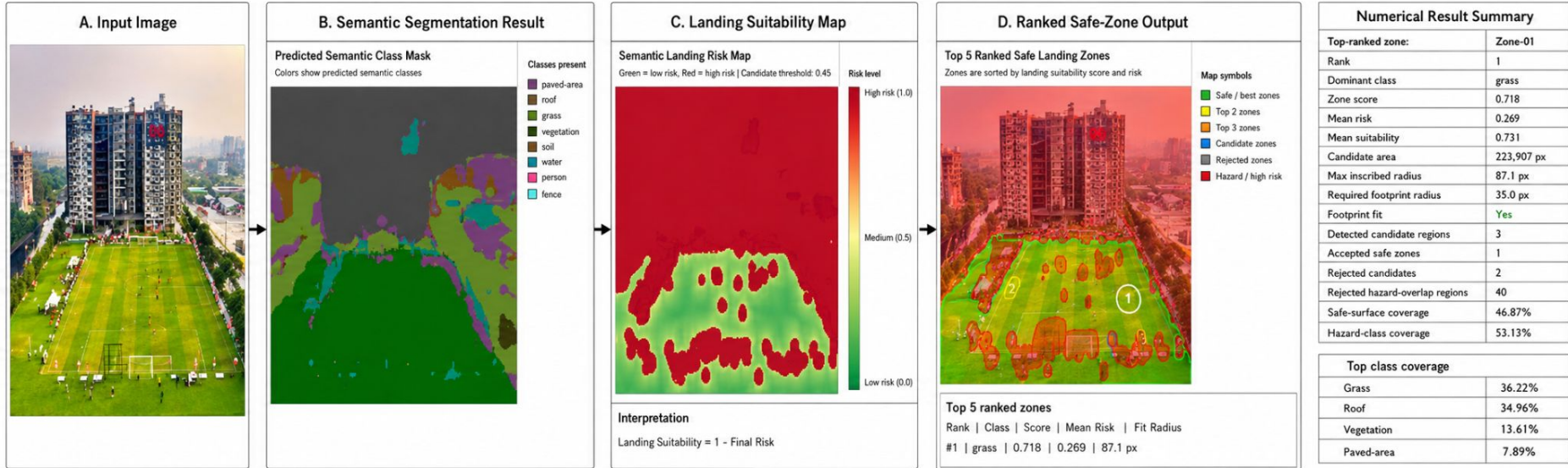
| Class            | Water                                                                             | Vegetation                                                                        | Roof                                                                              | Wall                                                                              | Fence                                                                             | Person                                                                             | Dog                                                                                 | Car                                                                                 | Bicycle                                                                             | Tree                                                                                | Obstacle                                                                            |
|------------------|-----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|
| Sample Image     |  |  |  |  |  |  |  |  |  |  |  |
| Risk Level       | HIGH                                                                              | HIGH                                                                              | HIGH                                                                              | HIGH                                                                              | HIGH                                                                              | HIGH                                                                               | HIGH                                                                                | HIGH                                                                                | HIGH                                                                                | HIGH                                                                                | HIGH                                                                                |
| Risk Score (0–1) | 0.90 – 1.00                                                                       | 0.80 – 1.00                                                                       | 0.85 – 1.00                                                                       | 0.85 – 1.00                                                                       | 0.80 – 0.00                                                                       | 0.60 – 0.85                                                                        | 0.60 – 0.80                                                                         | 0.70 – 0.90                                                                         | 0.60 – 0.85                                                                         | 0.60 – 0.85                                                                         | 0.90 – 1.00                                                                         |

# Post-Processing

The segmentation result is refined using:



# Outcome



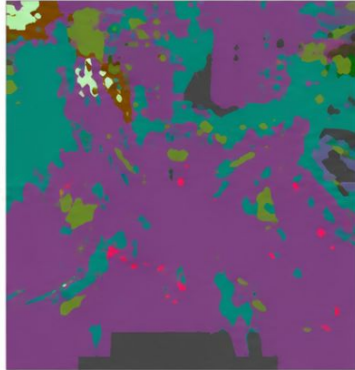
Sample Image: UIU Field

# Outcome

A. Input Image

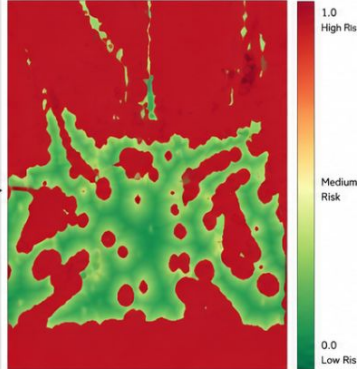


B. Semantic Segmentation Result



■ paved-area ■ dirt ■ grass ■ gravel ■ vegetation  
■ roof ■ wall ■ person ■ bold-tree ■ obstacle

C. Landing Suitability Map



D. Ranked Safe-Zone Output



| Top ranked zone |            |       |           |            |
|-----------------|------------|-------|-----------|------------|
| Rank            | Class      | Score | Mean Risk | Fit Radius |
| #1              | paved-area | 0.743 | 0.200     | 86.3 px    |

Numerical Result Summary

## Zone summary

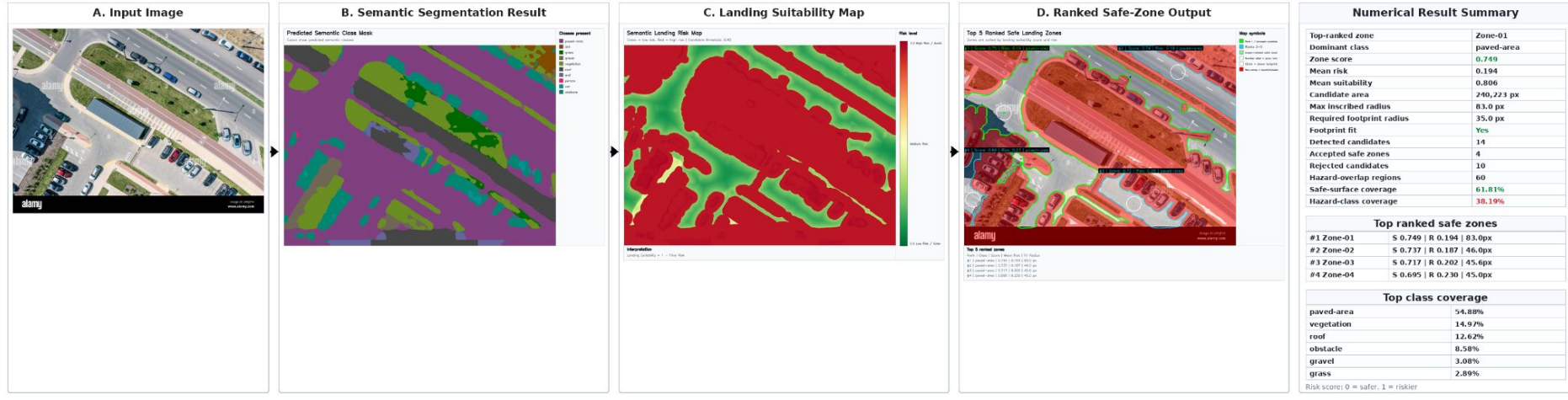
|                                  |            |
|----------------------------------|------------|
| Top-ranked zone:                 | Zone-01    |
| Rank:                            | 1          |
| Dominant class:                  | paved-area |
| Zone score:                      | 0.743      |
| Mean risk:                       | 0.200      |
| Mean suitability:                | 0.800      |
| Candidate area:                  | 326,115 px |
| Max inscribed radius:            | 86.3 px    |
| Required footprint radius:       | 35.0 px    |
| Footprint fit:                   | Yes        |
| Detected candidate regions:      | 18         |
| Accepted safe zones:             | 1          |
| Rejected candidates:             | 17         |
| Rejected hazard-overlap regions: | 64         |
| Safe-surface coverage:           | 71.10%     |
| Hazard-class coverage:           | 28.90%     |

## Top class coverage

|             |        |
|-------------|--------|
| Paved-area: | 67.70% |
| Obstacle:   | 17.97% |
| Vegetation: | 6.03%  |
| Roof:       | 3.46%  |

Sample Image: Gulshan 2 Circle

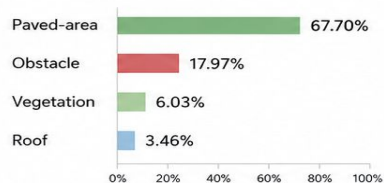
# Outcome



Sample Image: Road Side Top View

# Performance Metrics

## Semantic prediction quality



Prediction is dominated by paved-area, consistent with an urban road-intersection scene.

## Safe / unsafe separation



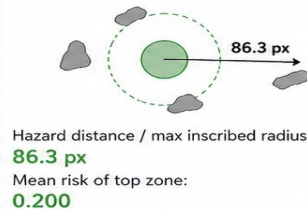
The suitability analysis separates safer paved regions from obstacle-dominated areas.

## Candidate zone area



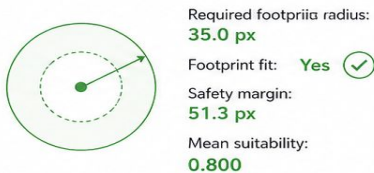
Many candidate regions are detected, but only one satisfies all landing constraints.

## Hazard distance



The selected zone maintains meaningful clearance from nearby hazards.

## Safety margin



The selected landing zone comfortably exceeds the minimum footprint requirement.

## Rejected region analysis



Post-processing aggressively removes unsafe or undersized regions.

## Ranked safe-zone interpretability

|                  |            |
|------------------|------------|
| Top-ranked zone: | Zone-01    |
| Dominant class:  | paved-area |
| Rank:            | 1          |
| Score:           | 0.743      |
| Mean risk:       | 0.200      |
| Risk category:   | Very Safe  |

Ranking improves explainability by reporting both semantic class and landing suitability score.

## Overall evaluation focus

- ✓ Semantic prediction quality
- ✓ Safe/unsafe separation
- ✓ Candidate zone area
- ✓ Hazard distance
- ✓ Safety margin
- ✓ Rejected region analysis
- ✓ Ranked safe-zone interpretability

Metrics derived from the generated class coverage, candidate zone, rejected region, and ranked safe-zone outputs.

Sample Image: UIU Field

# Limitations & Future Scope



## 1 RGB-Only Input

The system relies solely on RGB images. Absence of depth information limits accurate understanding of 3D structure and obstacle height.



## 2 Limited Terrain & Environmental Modeling

The system does not incorporate slope, roughness, or wind information, which can influence landing safety and feasibility assessment.



## 3 No Real Flight Validation

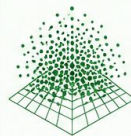
The approach has not been validated through real UAV experiments. Performance is evaluated only on dataset and simulation-based results.

### Limitations



## 1 RGB-D Integration

Incorporate depth information to improve obstacle detection, height estimation, and landing surface understanding.



## 2 3D Point Cloud & Terrain Analysis

Utilize 3D point clouds to estimate terrain slope and roughness for more reliable and risk-aware landing zone selection.



## 3 ROS2/Gazebo Simulation & Controller Integration

Develop ROS2/Gazebo simulation and integrate with a drone controller to enable autonomous landing and closed-loop testing.

### Future Scope

# Reference

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# Thank You

